TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

20260813_163108_7f4139

Date

2026-08-13T15:31:08.631938+00:00

Executive Summary

20260813_163108_7f4139

12	**6**	**0**
Total	Passed	Failed
6	**0**	
Skipped	Cancelled	

Overall measured availability: 73.15%

During this campaign, 6 of 12 scenarios passed, 0 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	1	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	1.813	11.297	True	True	8.222	21.435	10	16.67	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate real node	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	10.247	23.741	N/A	100.0	PASS	N/A	N/A
memory	swarm	N/A	N/A	N/A	Not Required	12.076	23.543	N/A	100.0	PASS	N/A	N/A
network-partition	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is	N/A
latency	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured	N/A
container-kill	kubernetes	6.391	19.422	True	True	1361.794	2042.522	7	22.22	PASS	N/A	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL	N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	24.089	60.518	N/A	100.0	PASS	N/A	N/A
memory	kubernetes	N/A	N/A	N/A	Not Required	17.887	29.536	N/A	100.0	PASS	N/A	N/A
network-partition	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured /renters=public: HTTP 401, /rentals=public:HTTP 401	N/A
latency	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured Observed: /cars=public:HTTP 401, /renters=public: HTTP 401, /rentals=public:HTTP 401	N/A

swarm — container-kill

Status: PASS

Detection Time (s): 1.813

Recovery Time (s): 11.297

HTTP Success: N/A

HTTP Failures: 10

Availability (%): 16.67

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

Status: PASS

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

Status: PASS

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

swarm — latency

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — container-kill

Status: PASS

Detection Time (s): 6.391

Recovery Time (s): 19.422

HTTP Success: N/A

HTTP Failures: 7

Availability (%): 22.22

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

Status: PASS

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — memory

Status: PASS

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — network-partition

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — latency

Status: SKIPPED

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

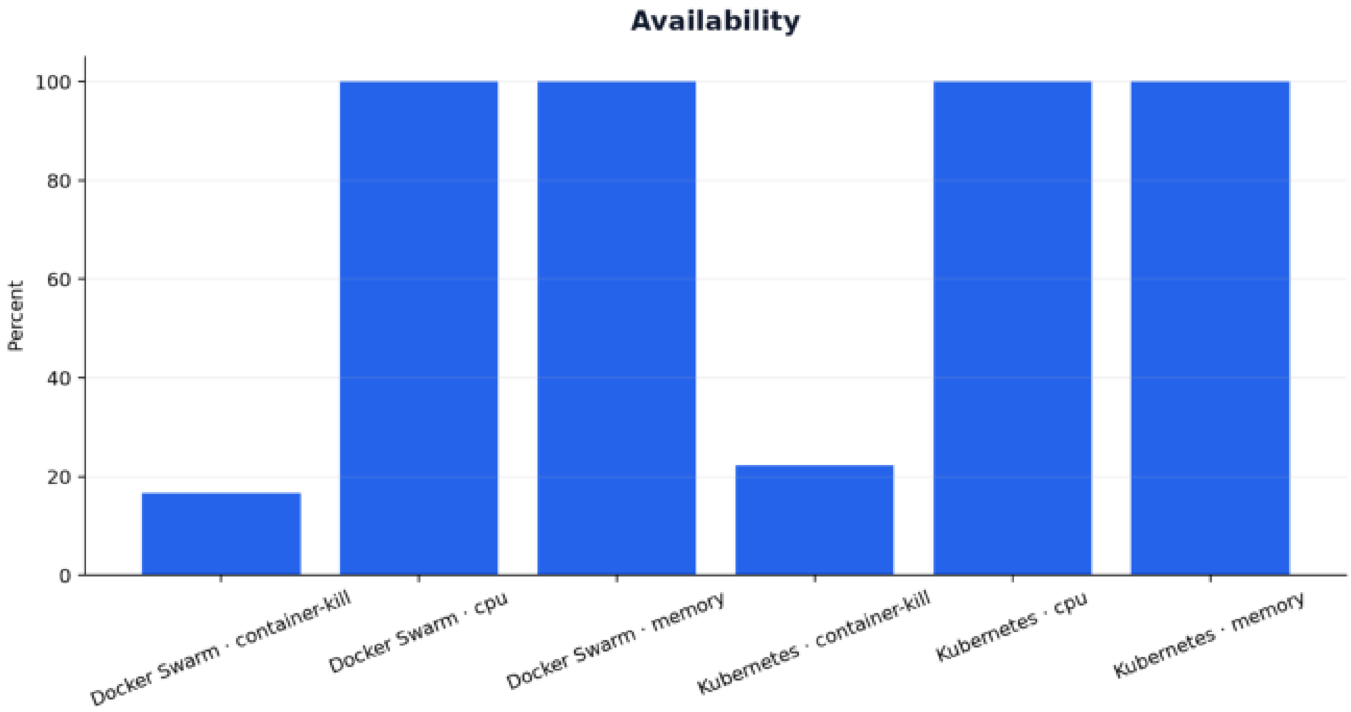
Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

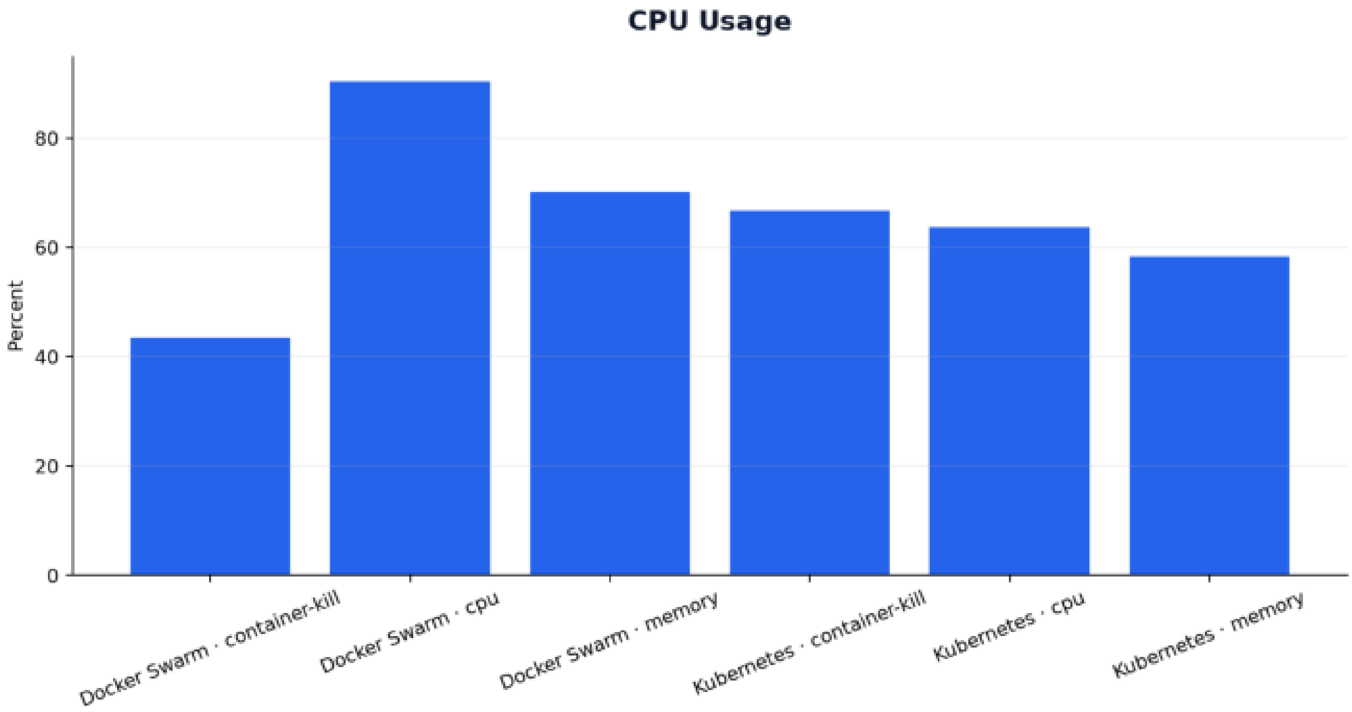
Availability

Current-session measured data



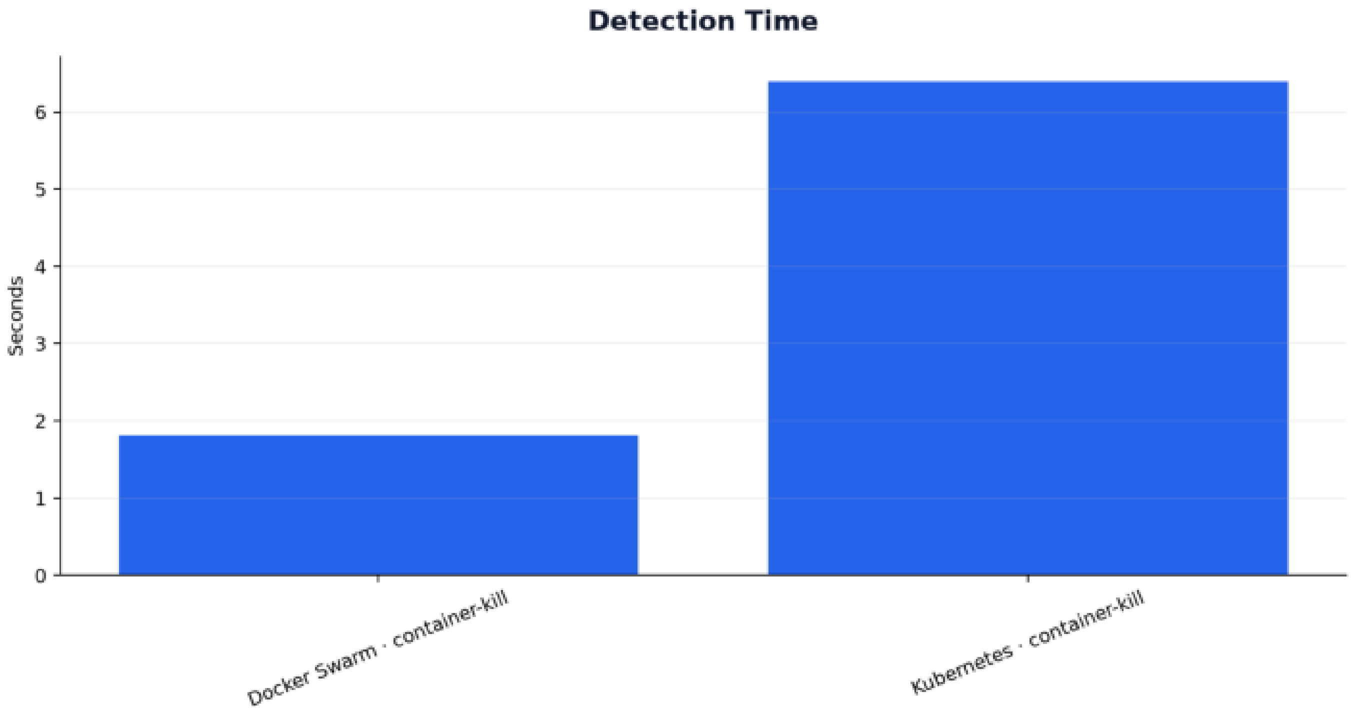
Cpu Usage

Current-session measured data



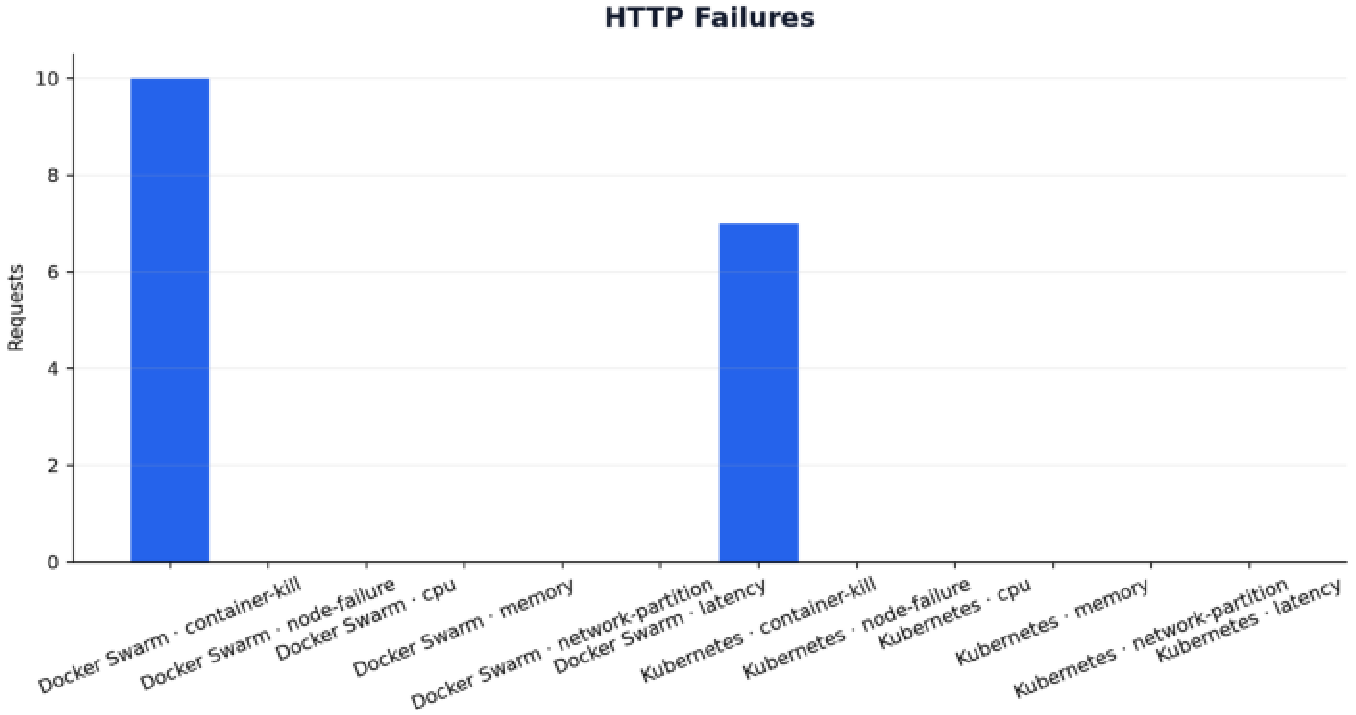
Detection Time

Current-session measured data



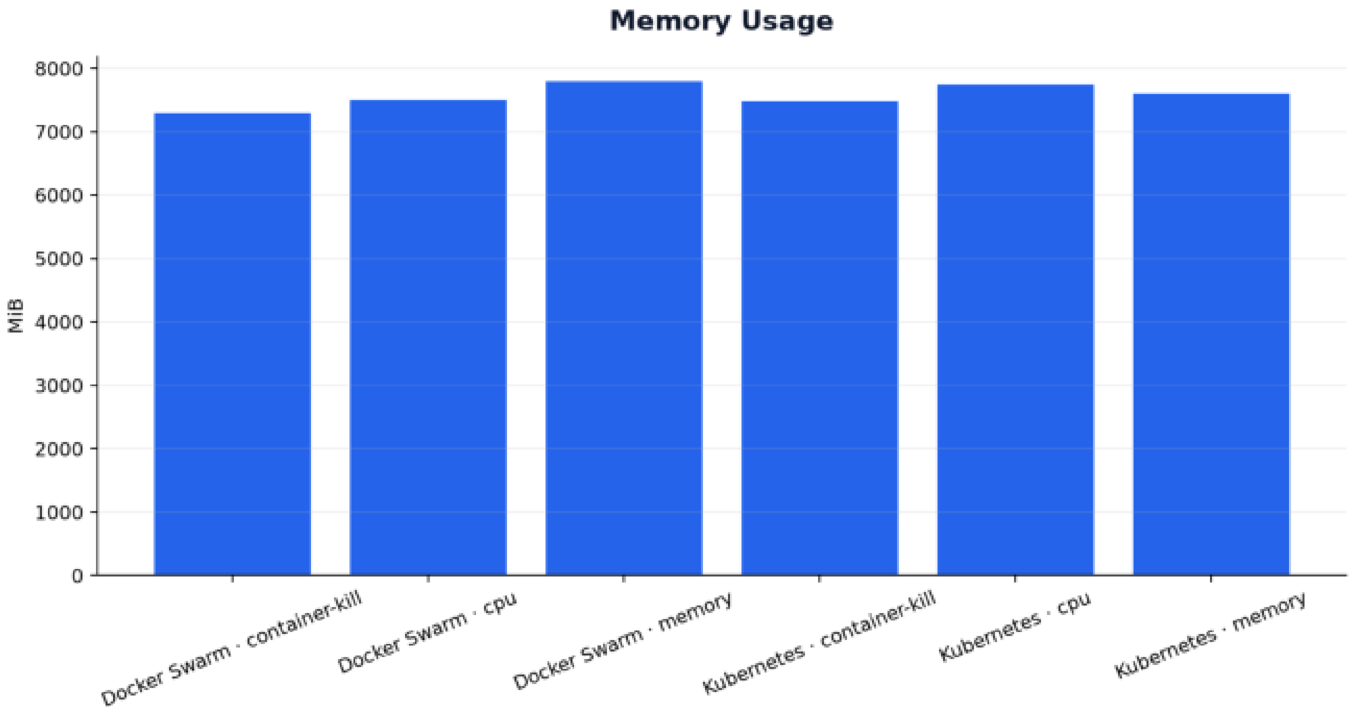
Http Failures

Current-session measured data



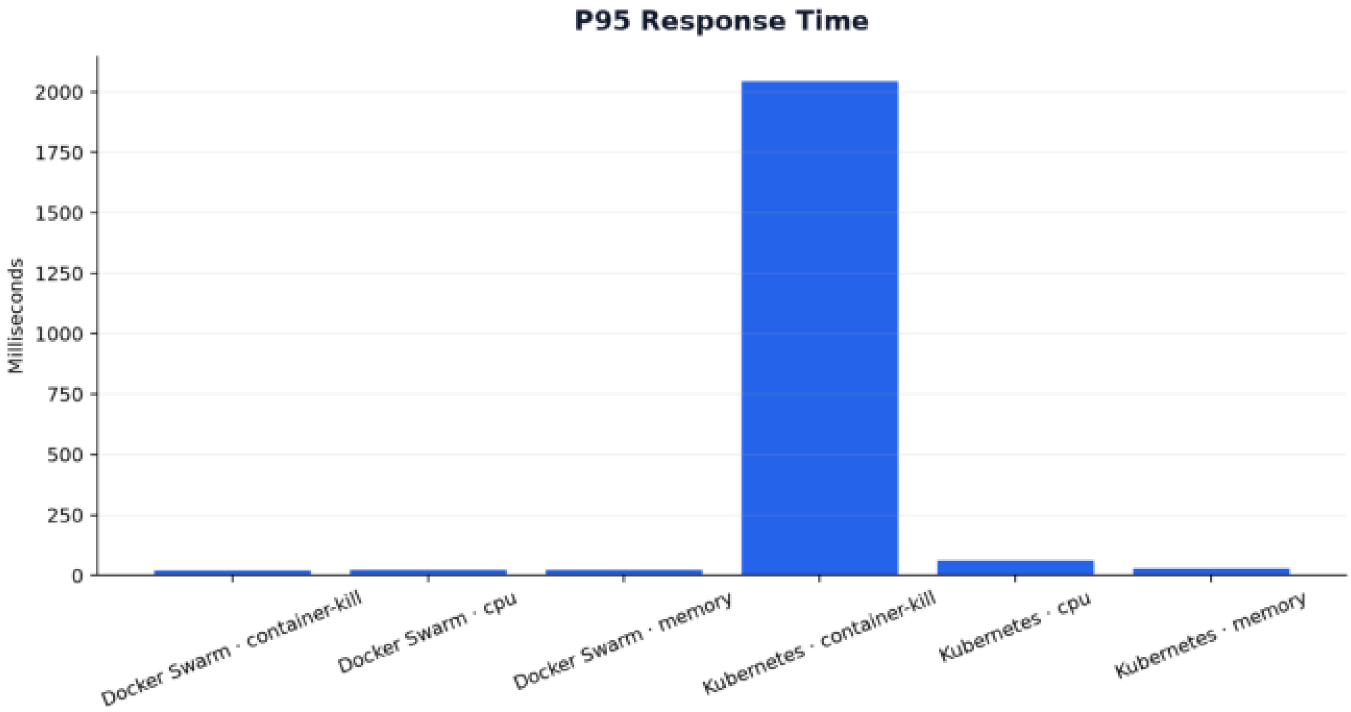
Memory Usage

Current-session measured data



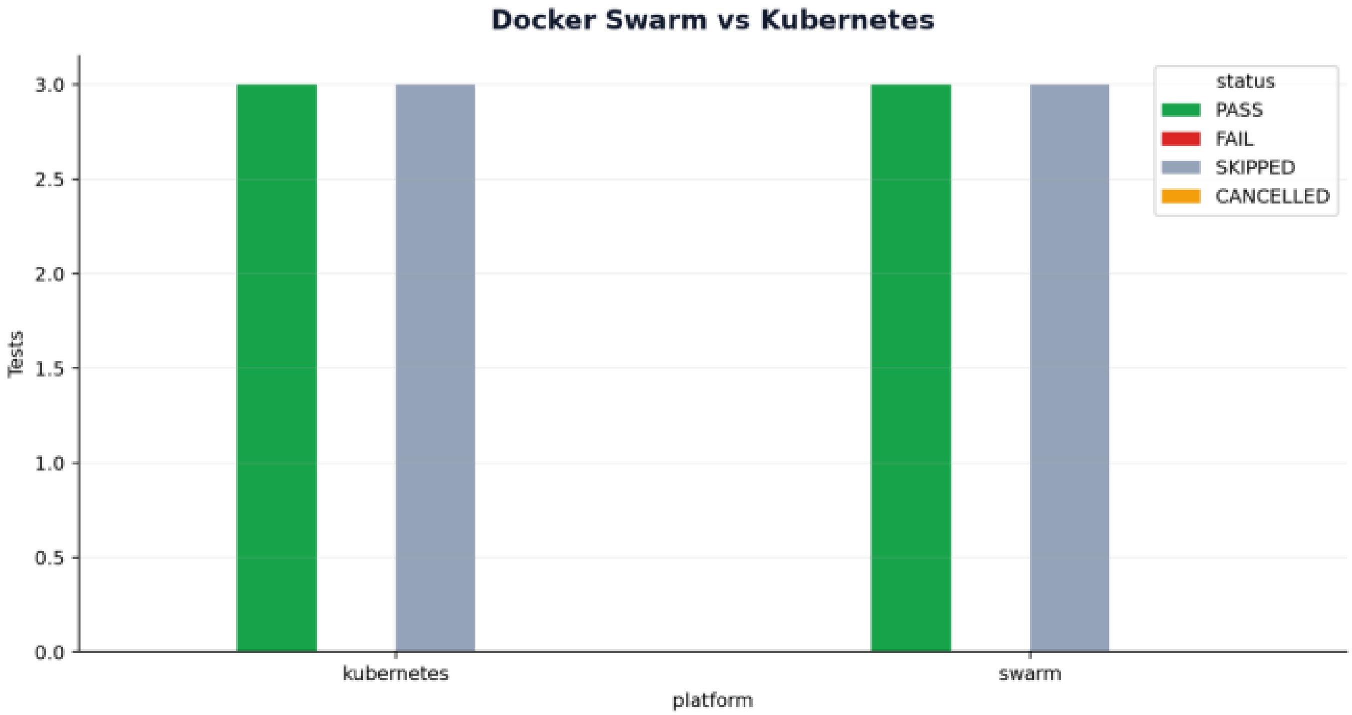
P95 Response Time

Current-session measured data



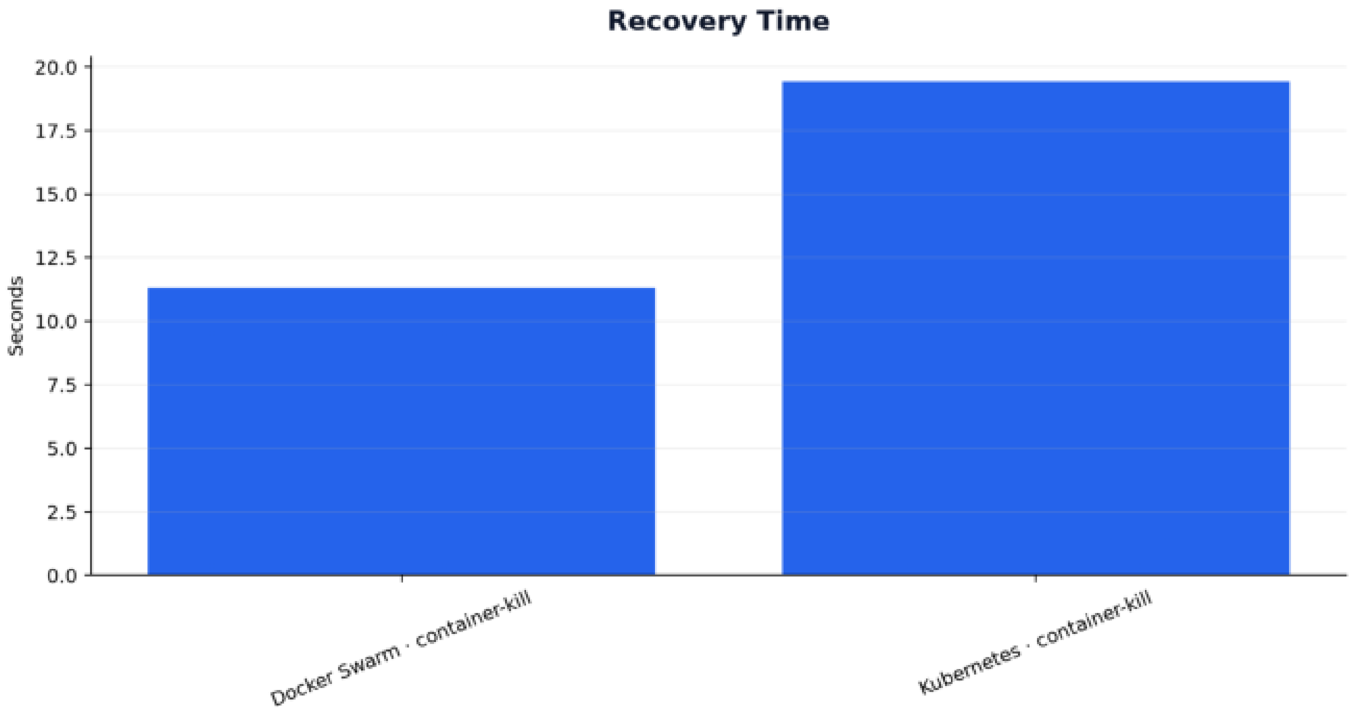
Platform Comparison

Current-session measured data



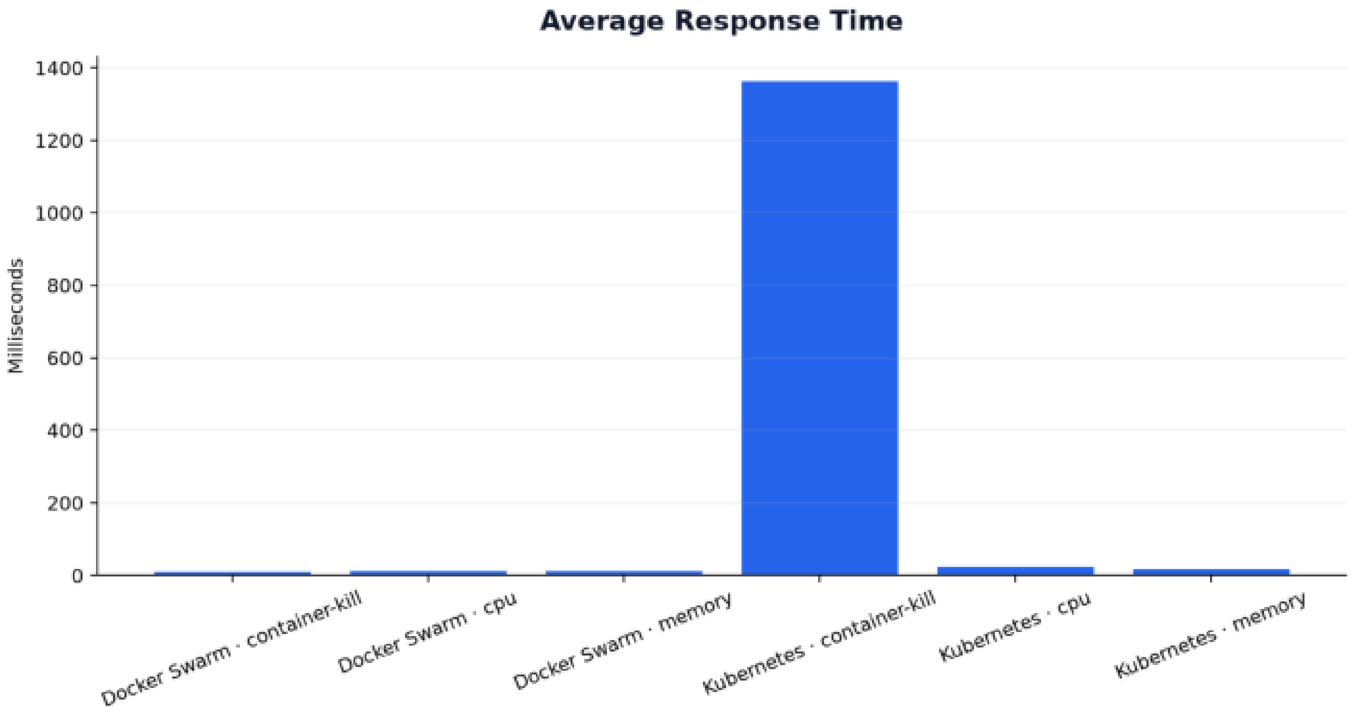
Recovery Time

Current-session measured data



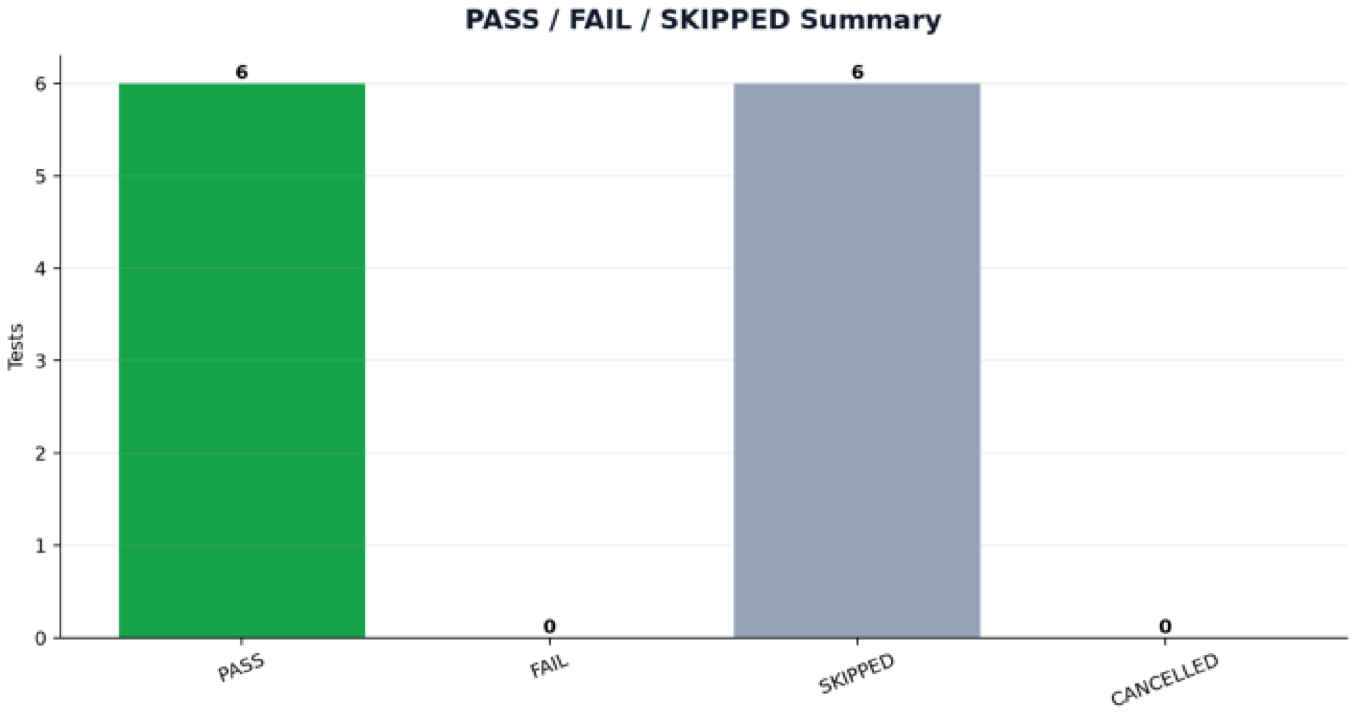
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
—	—	—	No failures recorded.

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling.
swarm	network-partition	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIE CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSW is configured. Observed: /cars=public:HTTP
swarm	latency	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIE CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSW is configured. Observed: /cars=public:HTTP
kubernetes	node-failure	Minikube inter-node CNI/DNS does not provide a stable backend-to-PostgreSQL path after rescheduling.
kubernetes	network-partition	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIE CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSW is configured. Observed: /cars=public:HTTP
kubernetes	latency	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIE CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSW is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

Swarm vs Kubernetes Comparison

Platform	PASS	SKIPPED
kubernetes	3	3
swarm	3	3

Final Conclusion

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During this campaign, 6 of 12 scenarios passed, 0 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.